

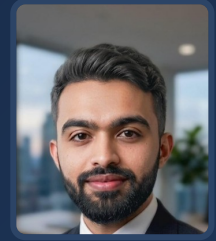
MEHDI BELHAJ

Power Electronics & Hardware Engineer

Analog & digital electronics · Embedded systems · Power conversion · Modelling & control — Aerospace · Space · Defence · Energy

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270 VDC · 540 V

SiC 1200 / 1700 V · 10 kW

DO-160G · DAL A

EMC / EMI

OrCAD · KiCad

Simulink · PSIM · LTspice

C · C++ · Python

AVAILABILITY

Immediate

EXPERIENCE

3 years · aerospace

CONTRACT

Permanent · Toulouse

WORK PERMIT

Valid to Nov. 2028

DRIVING LICENCE

B

ENGLISH

C1 · fluent

PROFILE

Power electronics and hardware engineer (Master's degree, 5-year) with 3 years of aerospace experience at Safran, Airbus and SuperGrid Institute. I design complete high-voltage embedded boards — Flyback 540 V, Buck, SEPIC and LDO supplies, three-phase SiC 1200 V inverter up to 10 kW, sensing chains, DO-160G EMC filtering — up to DAL A criticality. Sole electronics lead on my last two assignments, I cover the full V-cycle from specification to qualification, linking hardware design, simulation and control (Simulink / Simscape, PI / PID controllers).

KEY STRENGTHS

- **Complete hardware design** — analog, digital and power: from schematics to the PCB layout specification, including component selection and qualification.
- **EMC and aerospace qualification** — common- and differential-mode filters, lightning, ESD, DO-160G campaigns, insulation validated up to 20 kV.
- **Simulation, control and testing** — PSIM, LTspice, Simulink / Simscape, PI / PID controllers, then prototyping, debugging and bench testing.

SKILLS

POWER ELECTRONICS

Three-phase SiC 1200 V — 10 kW pulsed inverter · Quasi-resonant Flyback 540 V to 24 V without optocoupler · Isolated multi-output Flyback · Buck · SEPIC · LDO · Isolated gate drivers UCC21750 (DESAT, Miller clamp) · RCD snubbers · DC-bus pre-charge and inrush limiting · Magnetics (transformers, inductors, ferrites)

ANALOG & SENSING

Current sensing (ACS37002, INA190, shunts) · High-voltage sensing (resistive dividers, MAX1363 ADC) · Temperature (TMP1075) · 24-bit force measurement chain (ADS1247) · Position interface · Op-amps · Signal conditioning and filtering · Galvanic isolation

EMC, PROTECTIONS & TESTING

DO-160G Section 21 — cat. B and H: conducted and radiated emissions (LISN) · Common- and differential-mode filters · Lightning (pin injection) · 15 kV ESD · Phase-loss detection · Dielectric tests up to 20 kV · Insulation measurements · IEC 60068 · Bench bring-up and debugging, oscilloscope

CAD & PCB

OrCAD Capture CIS · KiCad · Proteus · IPC-2221 layout specification (altitude-corrected HV clearances, IEC 60664-1) · 6-layer FR4 PCB, stack-up · Current-based trace sizing · DFX and testability · Mechanical integration on 3D model

MODELLING & CONTROL

Matlab / Simulink / Simscape · Control System Toolbox · Simulink Control Design · Electrical and electromechanical equations · Block diagrams · Control-law design · PI / PID controller tuning · Closed-loop simulation · Stability (Bode, Nyquist, margins) · PSIM, LTspice, Saber

PROFESSIONAL EXPERIENCE



Power Electronics & Hardware Engineer, electronics lead — Safran Ventilation Systems

Nov. 2023 — May 2025 · 19 months

Via Centum T&S · Toulouse (Blagnac) · more-electric aircraft and helicopter · 270 VDC network · DO-160G category H

Input electronics of an aircraft ventilation system adapted to the 270 VDC high-voltage network — V-cycle covered autonomously.

- **Input-board design** (1,890 W, -40 to +100 °C): 9 functions designed and justified — reverse-polarity protection, isolated Flyback, LDO and Buck supplies, phase-loss detection in under 4 ms, micro-cut ride-through, inrush limiting, lightning protection, 15 kV ESD, EMC filter, thermal protection.
- **PCB design**: layout specification — 6-layer FR4 PCB, stack-up, IPC-2221 clearances corrected for altitude (IEC 60664-1), DC-bus traces rated for 9 A RMS, DFX and testability; mechanical integration on 3D model.
- **DO-160G cat. H EMC qualification**: common- and differential-mode filters, RCD snubbers, PSIM and FFT simulations; EMI test plans (LISN), lightning tests and insulation measurements; bench bring-up and debugging.
- **Documentation**: design justification file (calculation notes per function), hardware verification procedure, compliance matrix, lessons-learned sheet.
- **Coordination**: multidisciplinary meetings (hardware, layout, mechanical, procurement), internal and gate reviews, weekly technical meeting led in English with Safran USA.

Results: functional mock-up validated in EMC tests on the applicable DO-160 sections, complete design package delivered, 270 VDC feasibility demonstrated.

Motor-control electronics of an ultra-fast electromechanical actuator deploying a spoiler in under 130 ms: 3 boards within 65 × 65 × 170 mm.

- **Power boards:** 10 kW pulsed three-phase SiC inverter (6 × 1200 V MOSFETs, isolated UCC21750 drivers, DESAT); quasi-resonant Flyback 540 V to 24 V without optocoupler; Buck 24 V to 12 V.
- **CPU board** around a TMS320F280049 DSP (UART, SPI, I2C, RS422, 24-bit ADC).
- **Modelling and control:** electrical and electromechanical equations, Simulink / Simscape block-diagram models, control laws and controller tuning, closed-loop simulation, 130 ms requirement validated (Bode, Nyquist); LTspice, PSIM.
- **Coordination and reviews:** work with Airbus, Crouzet and Serma; design and reliability reviews; integration with mechanical engineers (flex circuits, 3D model).

Results: 130 ms dynamic requirement validated; hardware architecture approved in Airbus and Crouzet reviews.



Isolated high-voltage supply for the gate drivers of 6 series-connected 1.7 kV SiC MOSFETs in a high-voltage leg.

- **Design:** architecture with 12 cascaded ferrite transformers feeding the high- and low-side drivers; TDK EPCOS T38 core selection balancing losses, insulation and footprint.
- **Simulation, measurements and EMC:** LTspice modelling of the transformers' magnetic and electrical behaviour, validated on the oscilloscope; characterisation of common-mode currents generated by edges up to 16 kV/μs and countermeasures.
- **Testing and industrialisation:** prototyping, dielectric tests up to 20 kV to validate insulation and safety; specification for industrial prototyping (efficiency, EMC).

Results: 20 kV insulation validated, 12-transformer cascade operational, industrialisation specifications delivered to the prototyping team.

ENTREPRENEURSHIP — IN PARALLEL

European brand of electrical and sanitary home equipment, founded with four partners and sold online (Amazon, Shopify) in six countries. My role: product, compliance and operations. The business is now stable and run day to day by an employee.

- **Product compliance:** CE marking (Low Voltage and EMC directives), RoHS, WEEE, EPR, UKCA — laboratory test coordination (electrical safety, EMC), technical files and EU declarations of conformity with French and German bodies; German quality label "SEHR GUT (1.3)" (independent tests, 2025).

EDUCATION & CERTIFICATIONS



- **Python mini-project — scope-emi-toolkit:** oscilloscope CSV parsing, dBμV spectrum and EMC pre-check before test campaigns; 8 unit tests, continuous integration (github.com/mehdi-belhaj-eea).
- **C++ mini-project — embedded-dsp-cpp:** ring buffer and digital filters in C++17 for embedded targets; 12 GoogleTest tests.



LANGUAGES

French — bilingual · **Arabic** — native · **English** — C1, fluent (technical meetings and deliverables with Safran USA) · **Spanish, Portuguese** — basics

SOFT SKILLS & INTERESTS

Autonomous · rigorous · methodical · curious · team player · proactive · Competitive football · NQT mentoring · travel — 18 countries visited

References and detailed skills portfolio available on request · Updated — October 2026